

Distributed Quantum Computing

Massimo Equi

Aalto University

15 June 2026

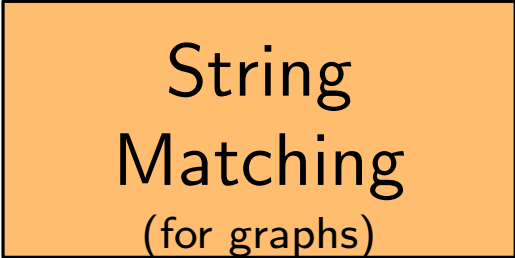
Interview for IQM

About Me

Three main research topics

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An orange rectangular box with a black border, containing the text 'String Matching (for graphs)'.

String
Matching
(for graphs)

About Me

Three main research topics

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Quantum
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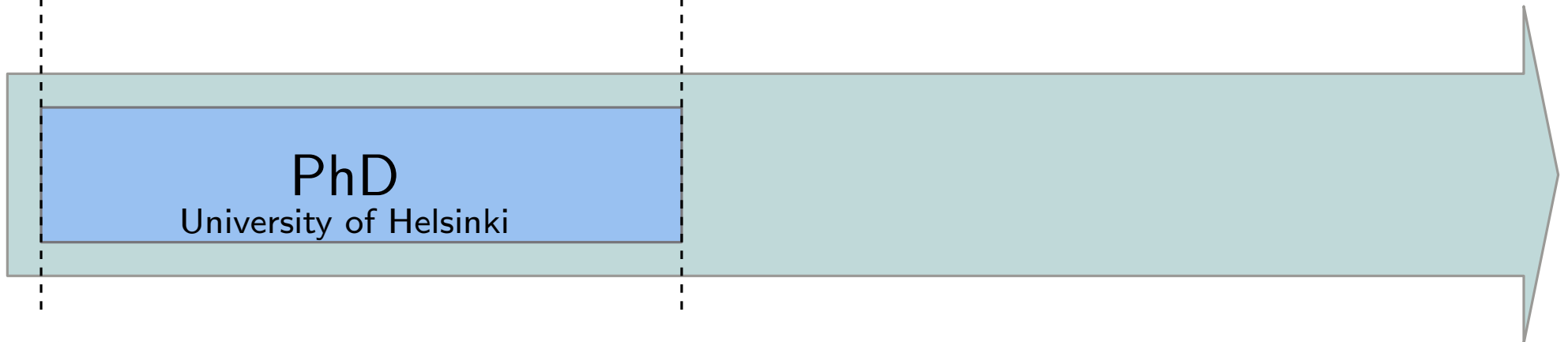
String Matching

PhD

University of Helsinki

2019

2022



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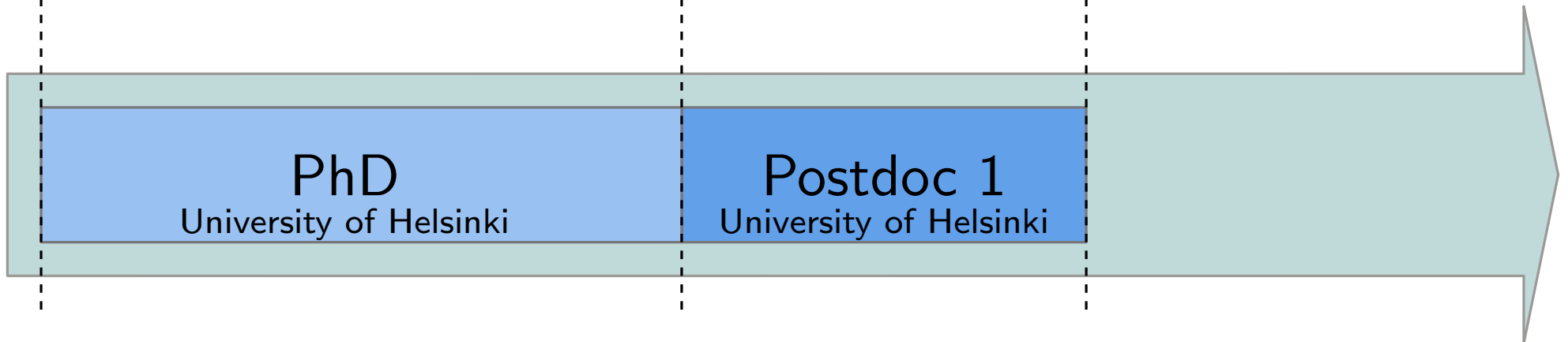
Postdoc 1

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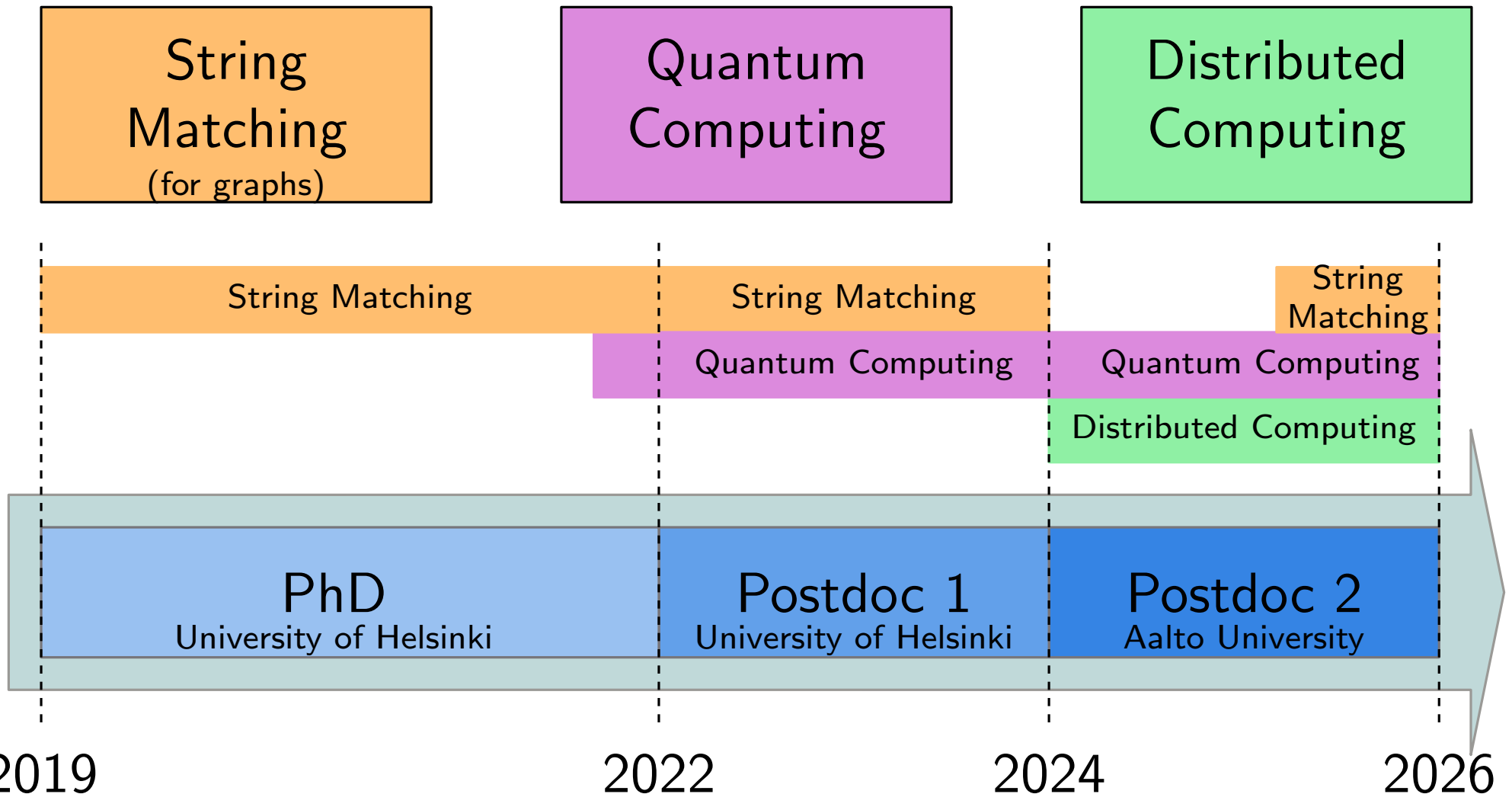
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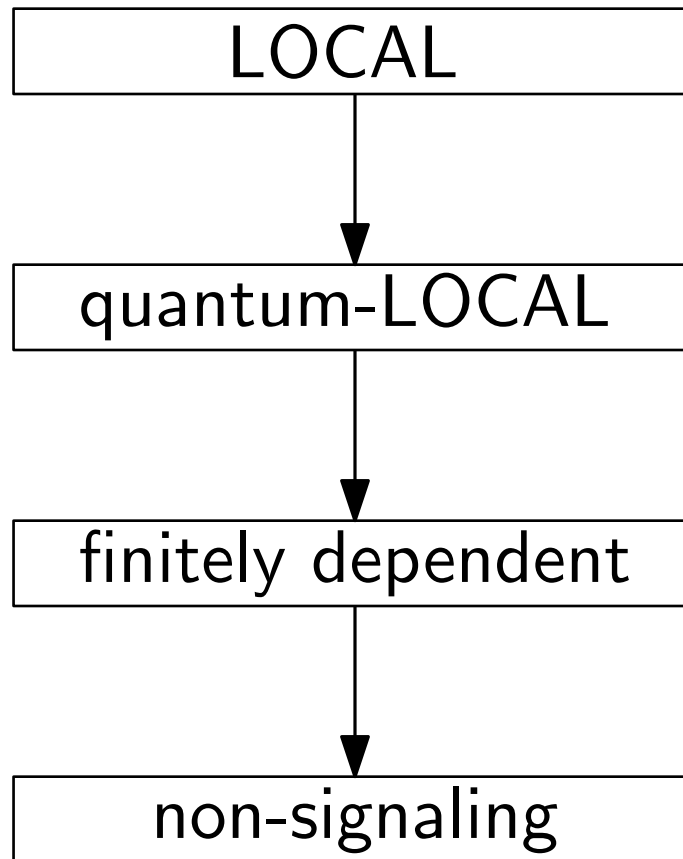
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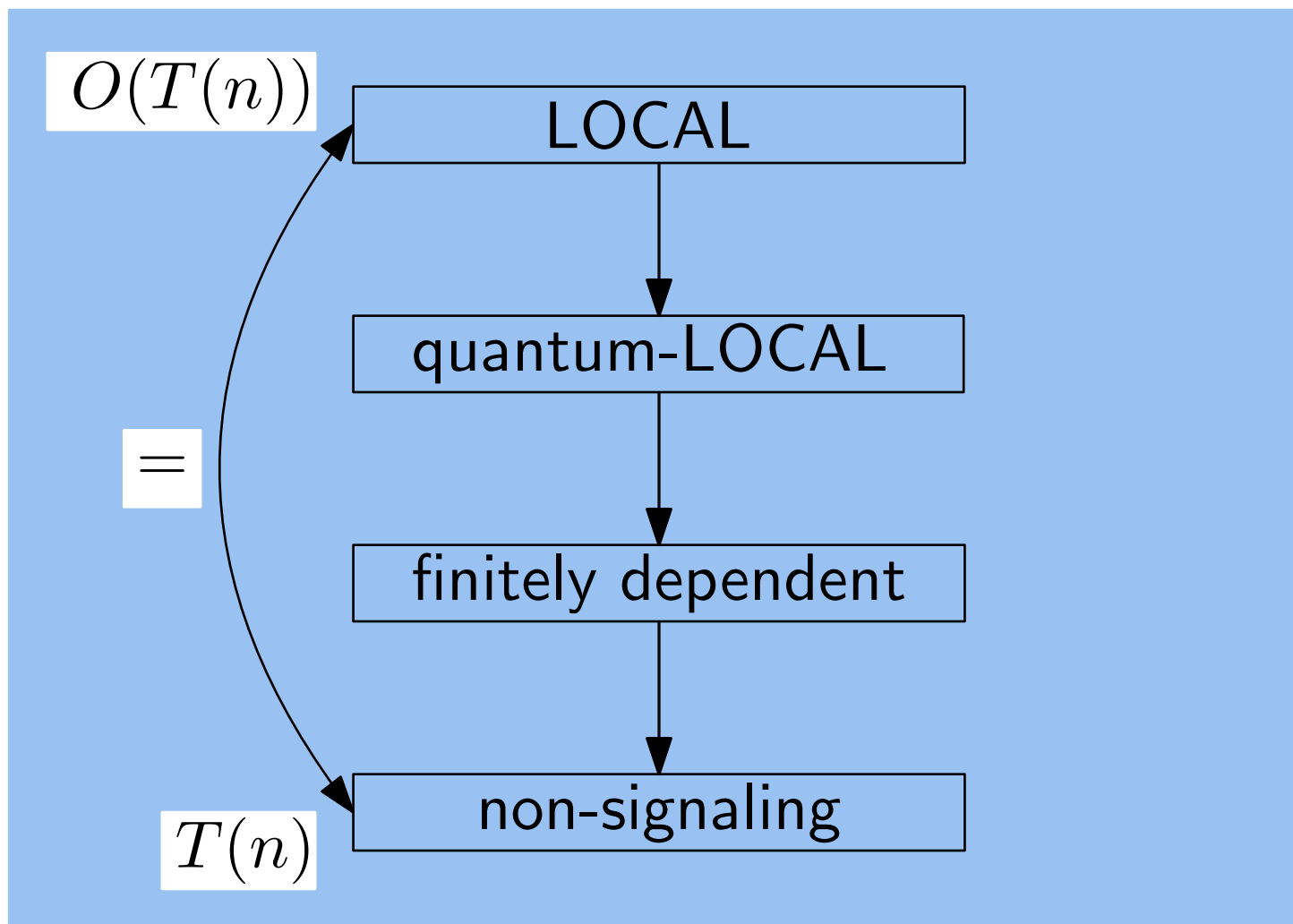


Distributed Hierarchy



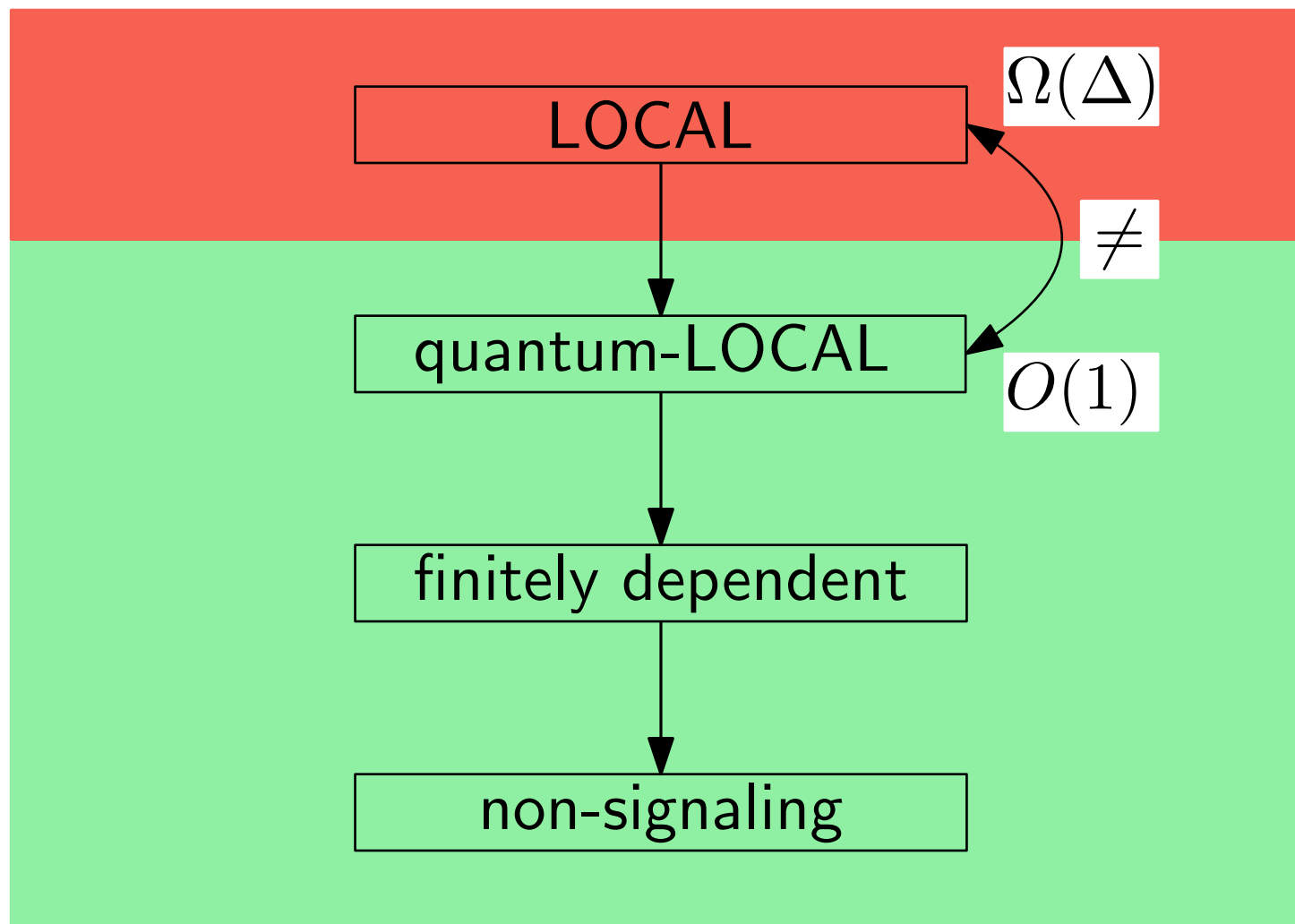
Distributed Hierarchy

For Linear Programs (LPs)

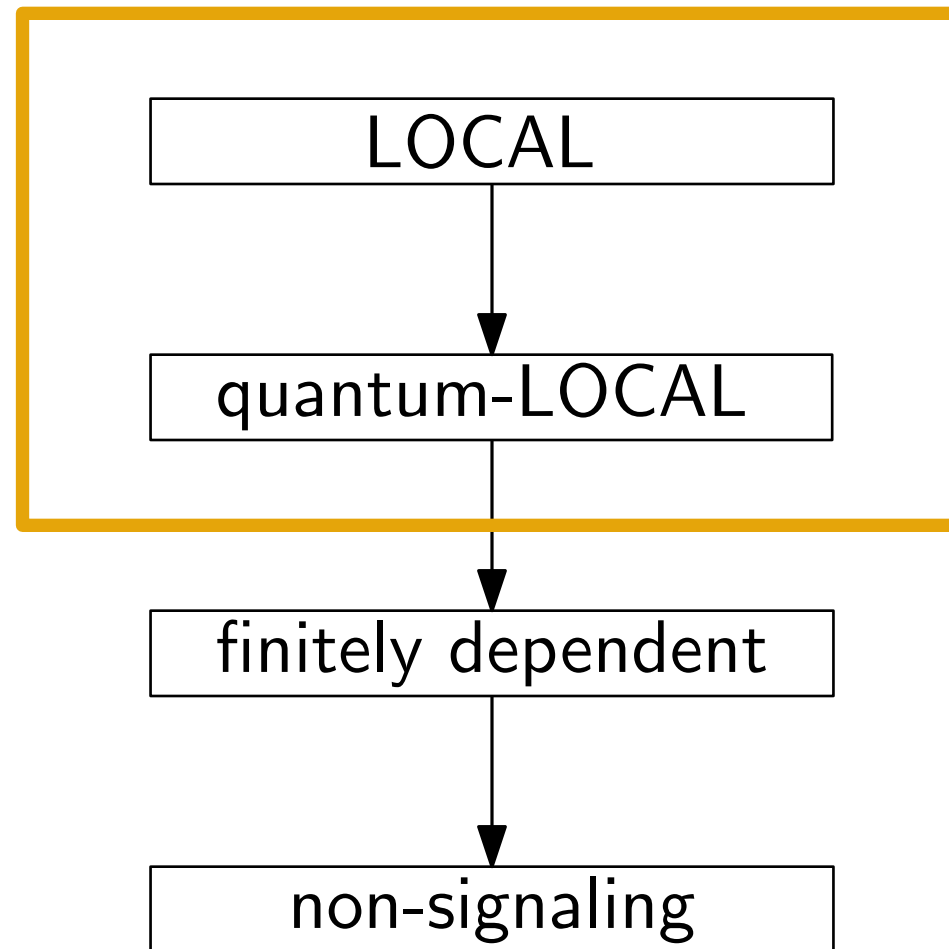


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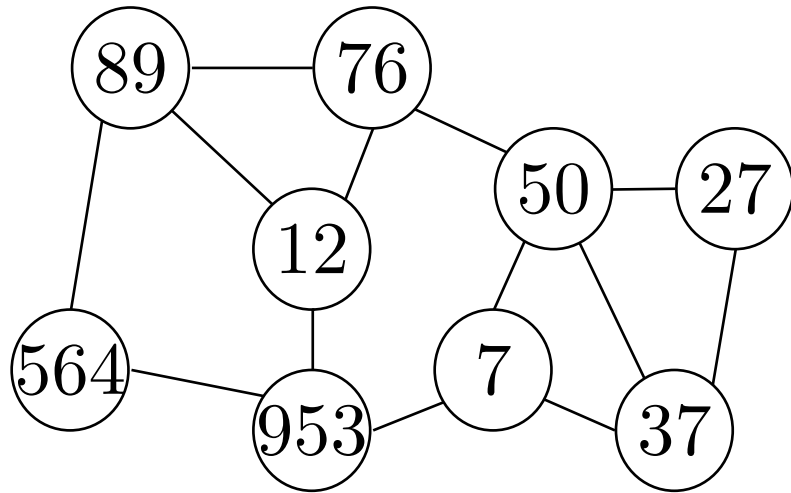
For Locally Checkable Labelling Problems (LCLs)



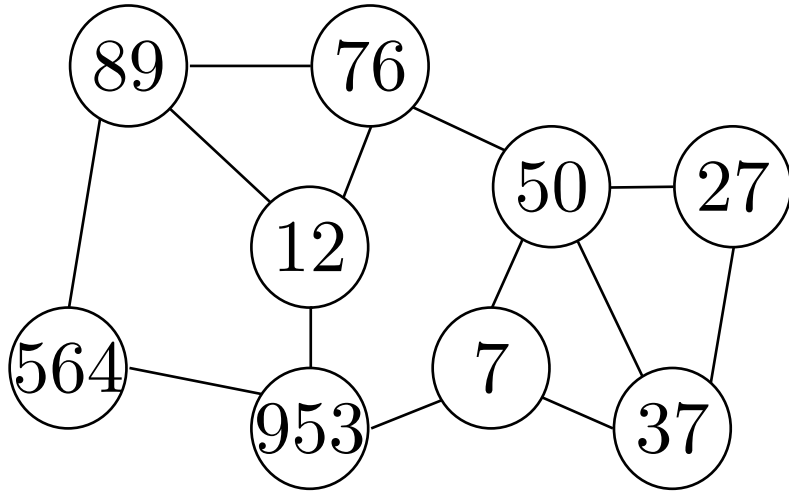
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LOCAL and quantum-LOCAL

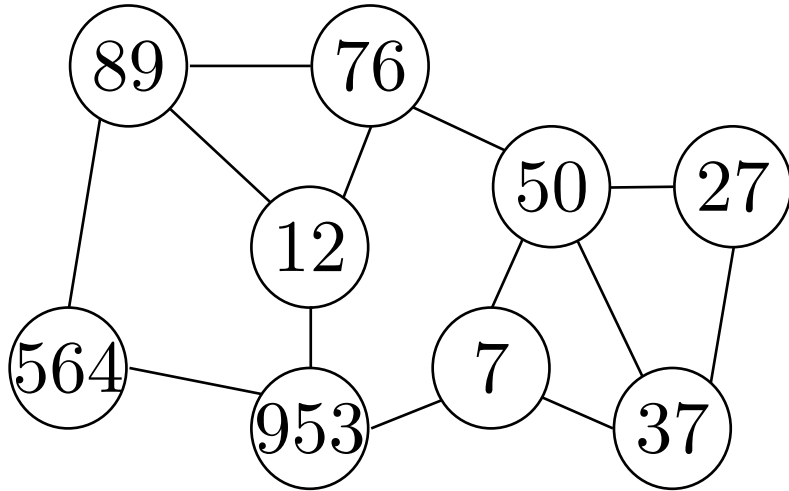


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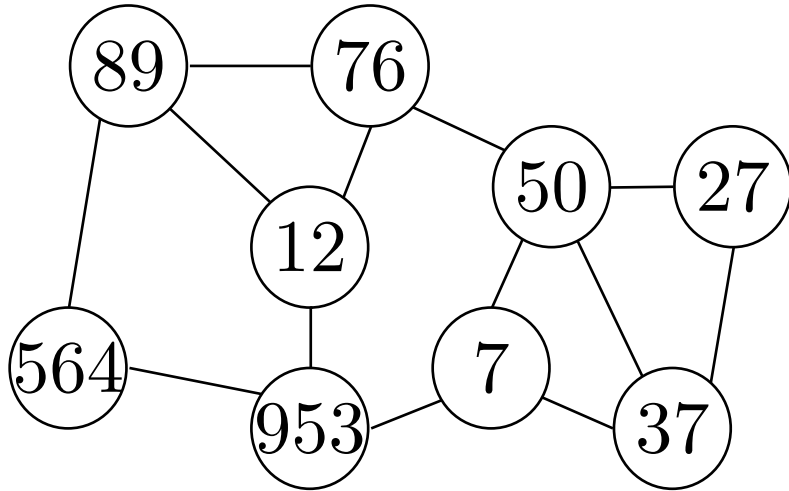
- The network is a graph $G = (V, E)$ of n nodes

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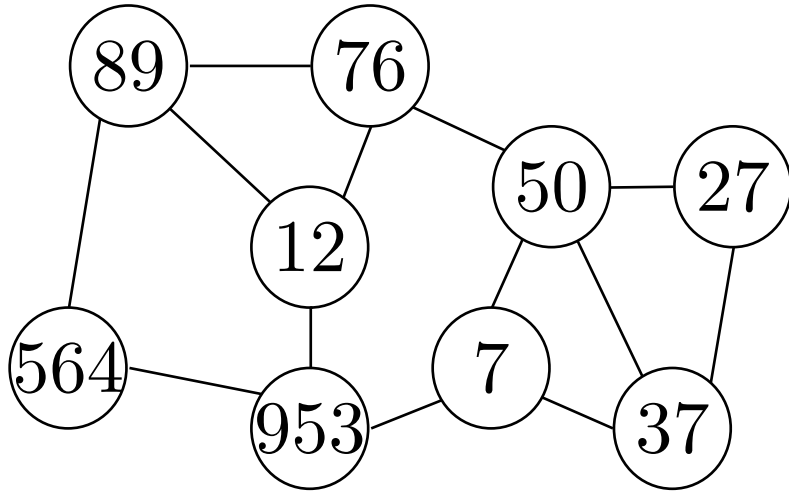
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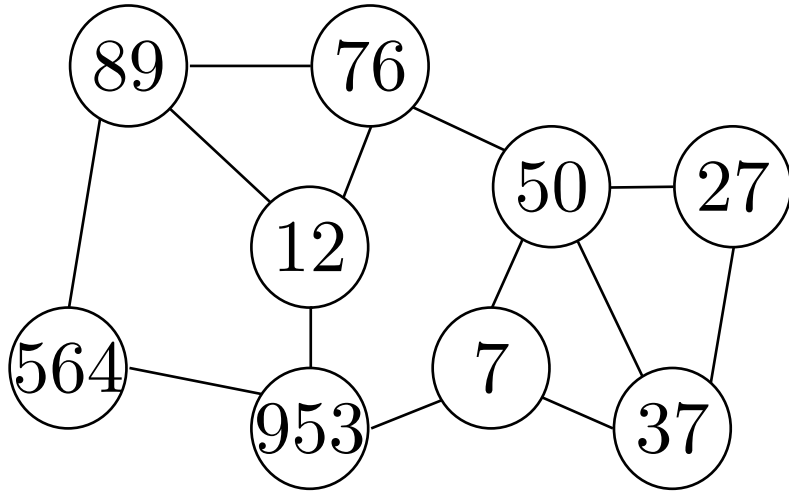
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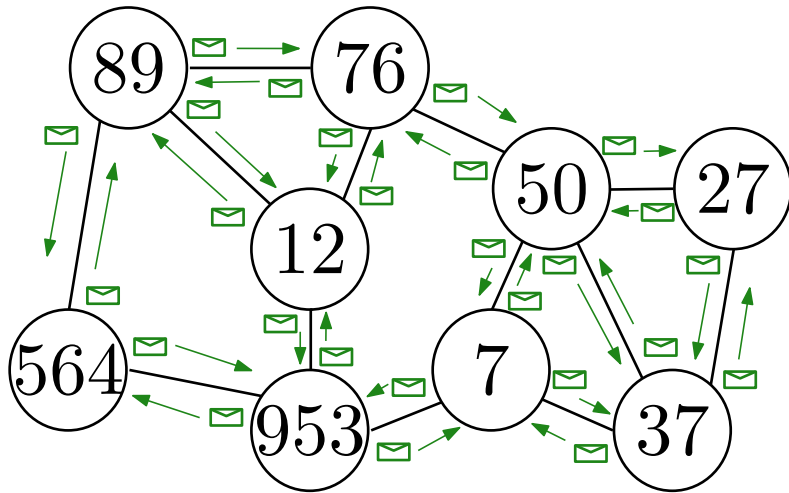
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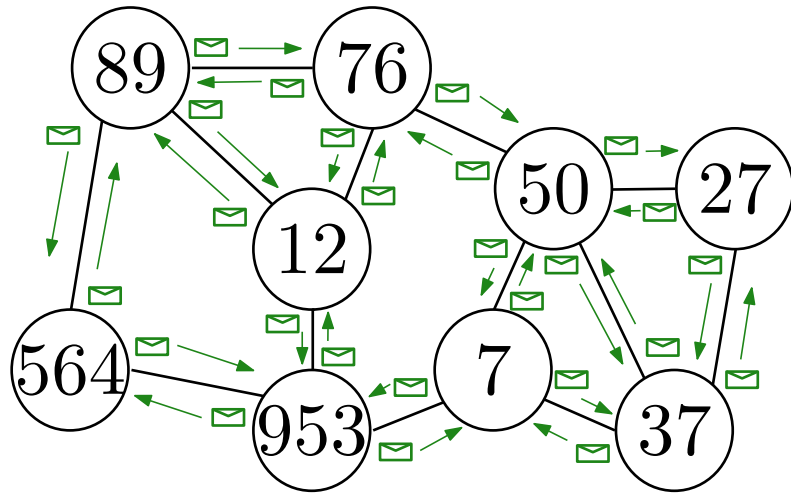
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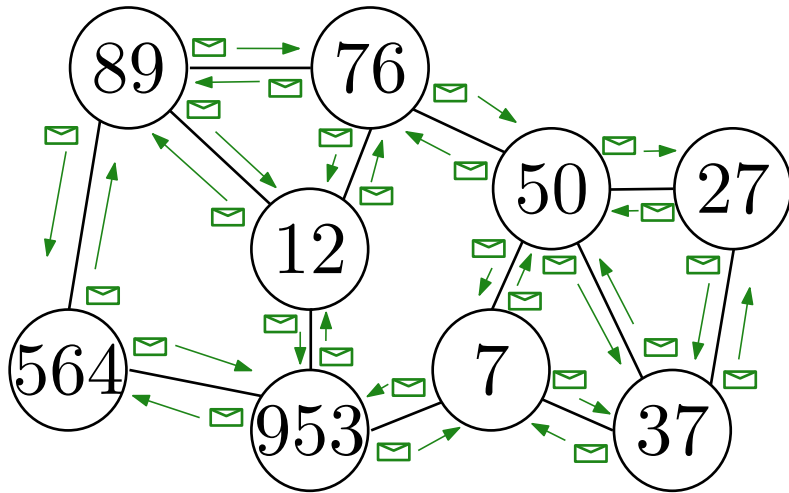


Complexity

Number of communication rounds
as a function of n

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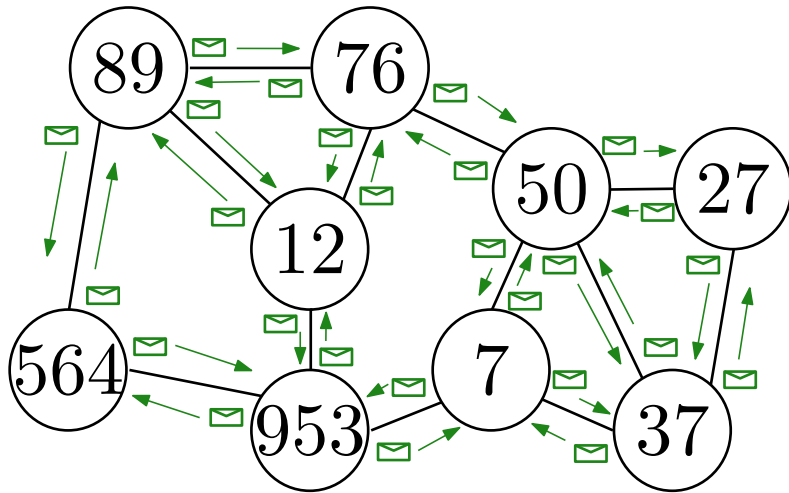
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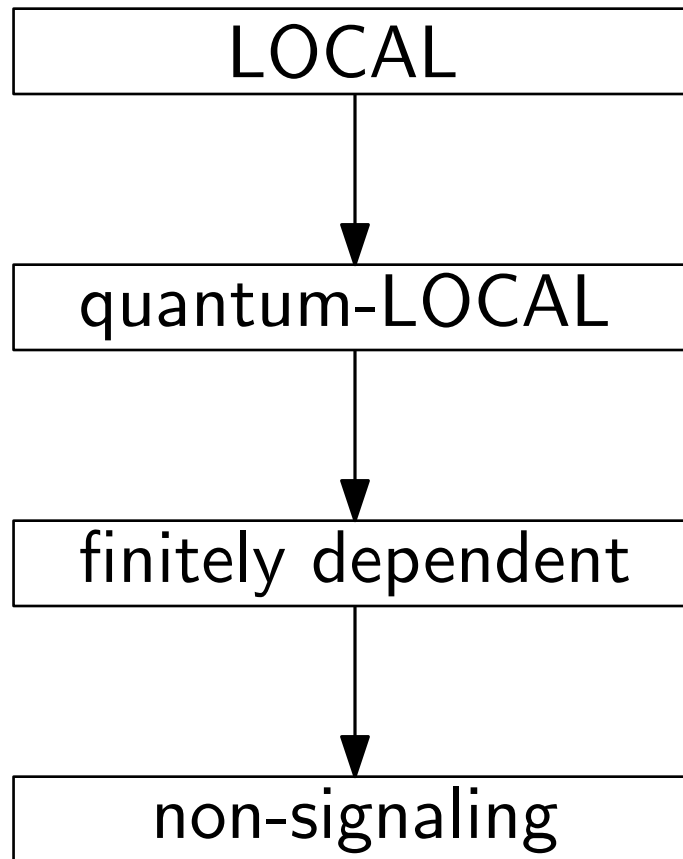
quantum-LOCAL

Nodes hold, send and receive **qubits**

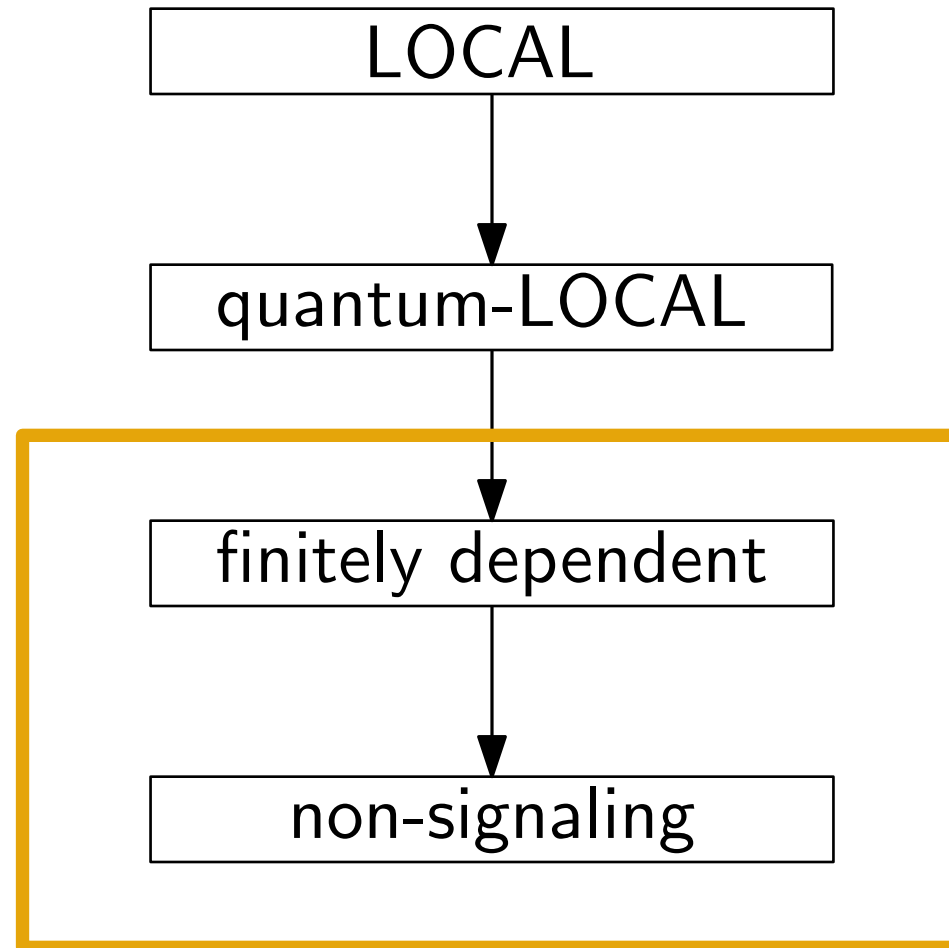
IDs are not needed

“send qubits” means applying
SWAP gates

Distributed Hierarchy

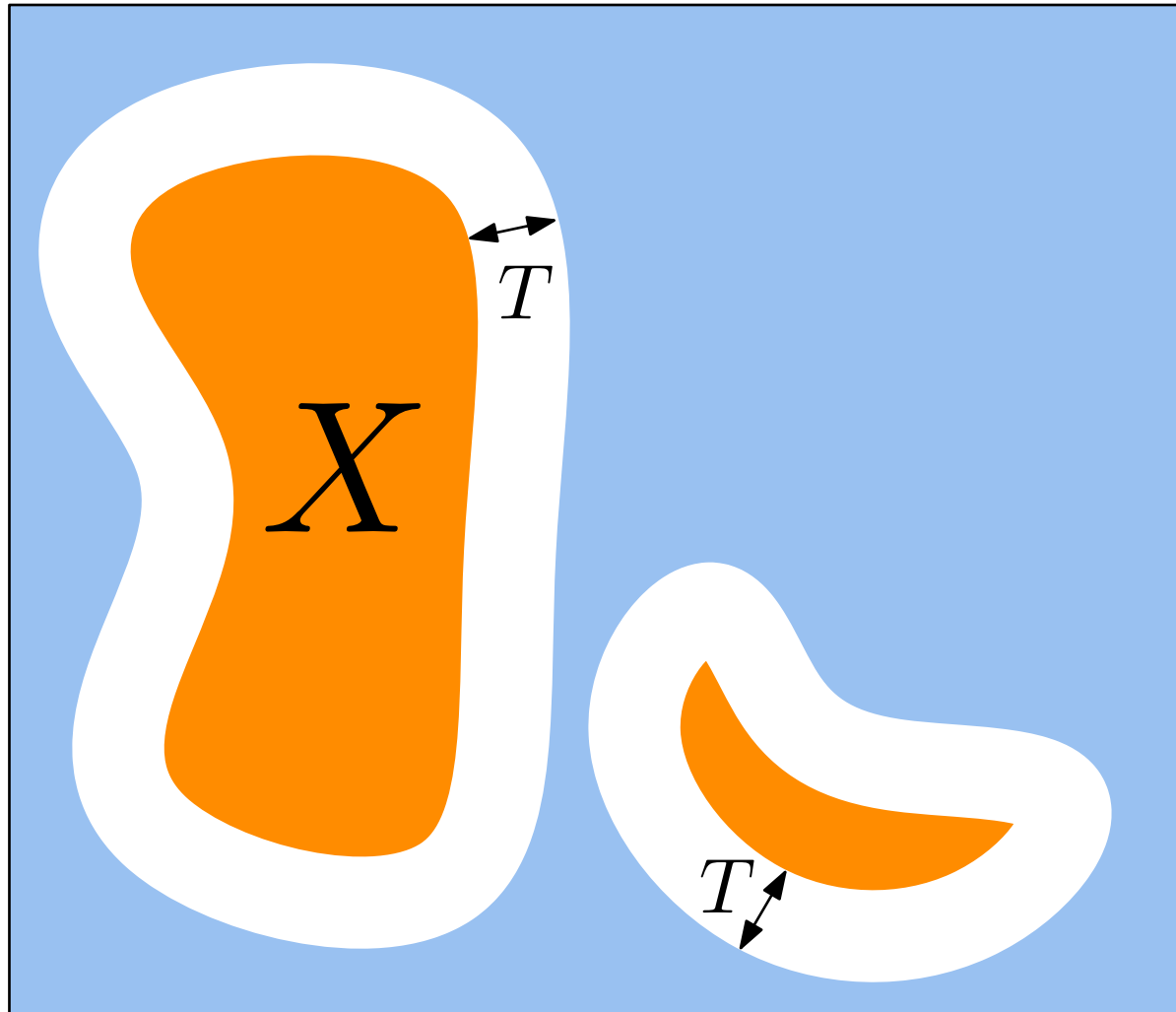


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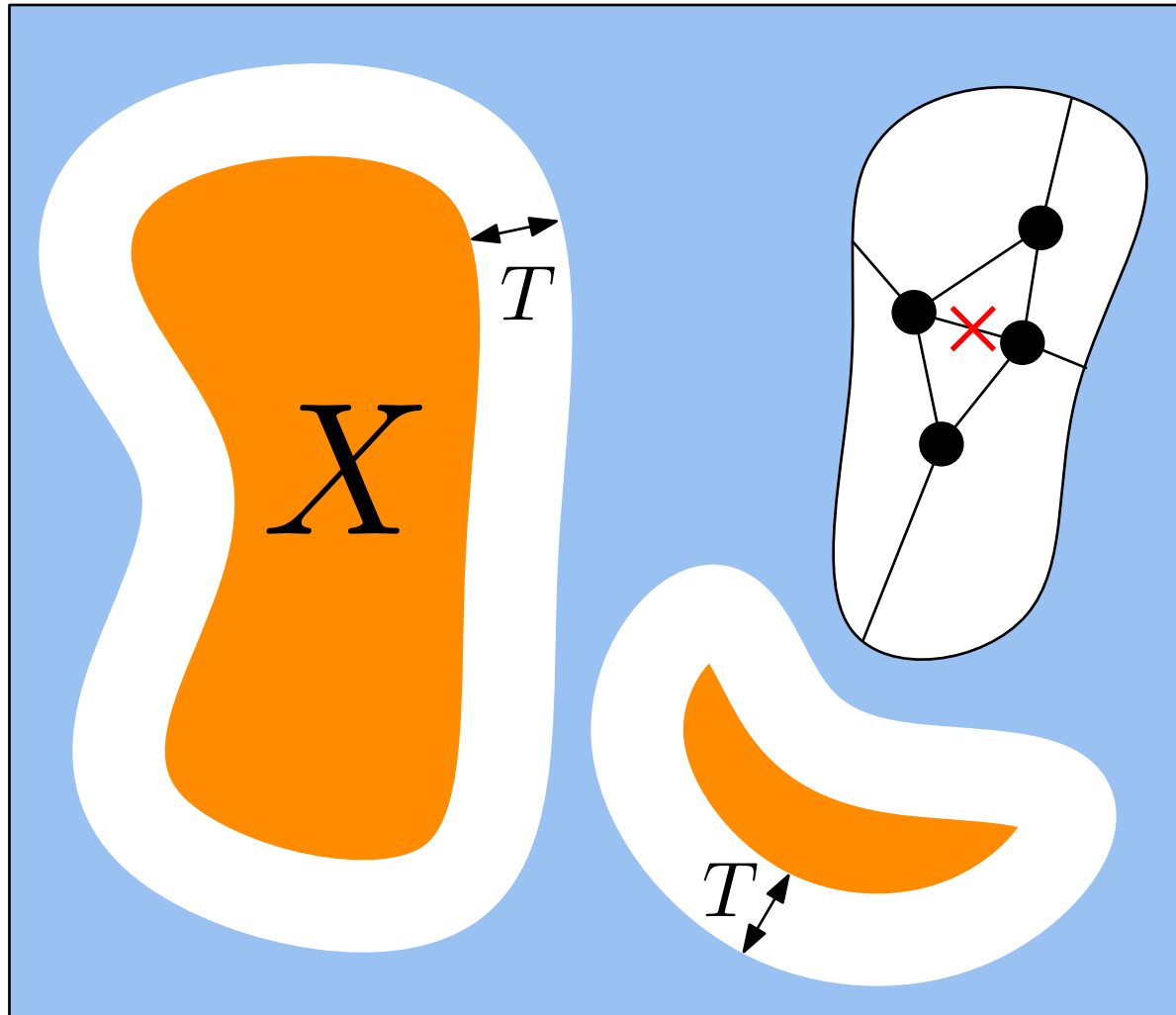
Non-signaling distributions

For any subset of nodes X , changes at distance $> T$
do not affect the output distribution of X



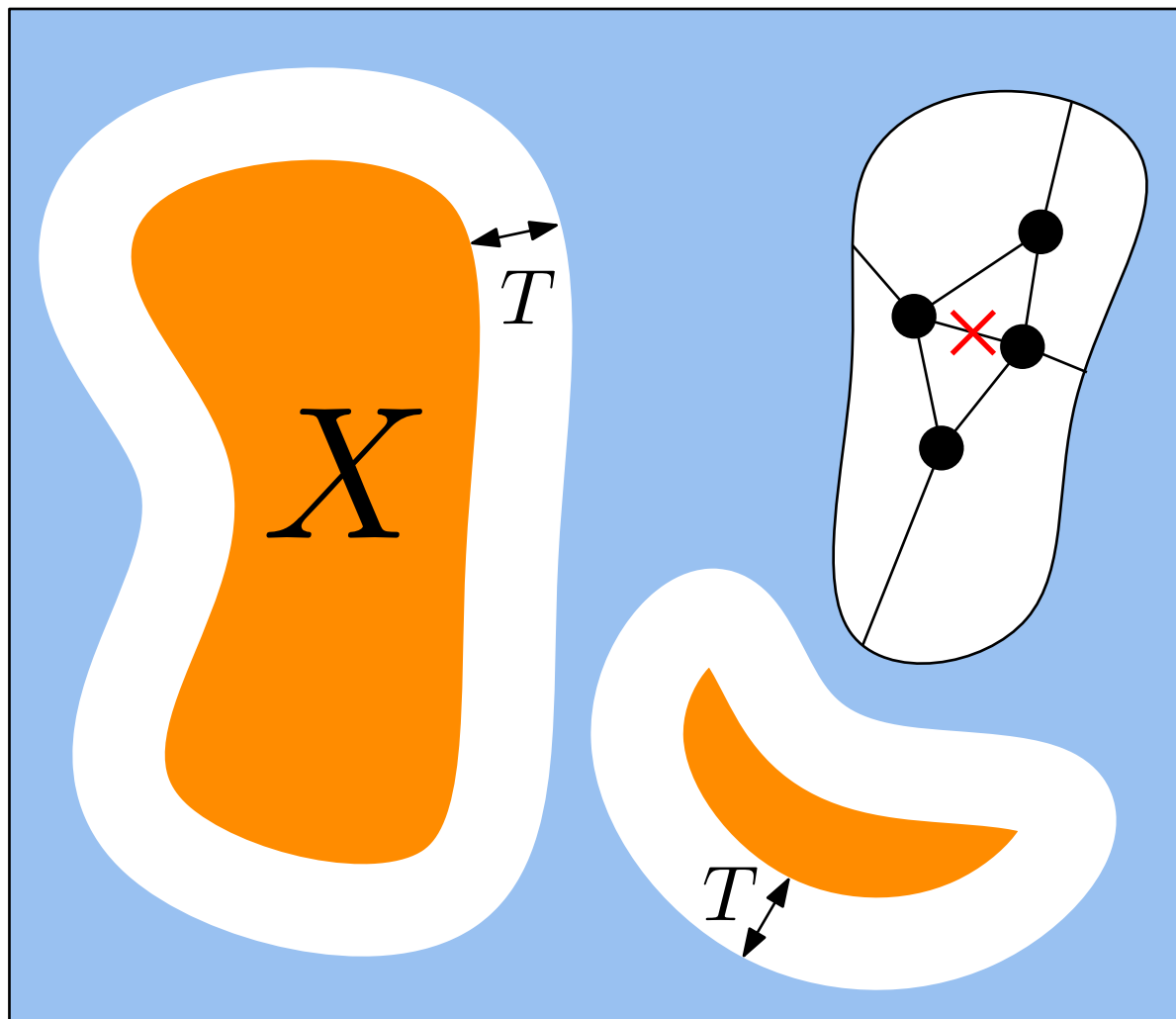
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Finetely dependent distributions add restrictions not considered today

Results for Linear Programs (LPs)

Reminder:

- **Independent Set (IS):**

a subset of nodes $I \subseteq V$ such that $\forall v, u \in I : (v, u) \notin E$

- **Maximal Independent Set (MIS):**

a subset of nodes $I \subseteq V$ such that $\forall u \in V \setminus I, \exists v \in I : (v, u) \in E$

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Example of Distributed LP: **fractional maximum independent set**

$$\text{maximize } \sum_{v \in V} x_v$$

$$\text{subject to } x_v + x_u \leq 1$$

$$0 \leq x_v \leq 1$$

$$\forall (v, u) \in E$$

$$\forall v \in V$$

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When $x_v \in \{0, 1\} \quad \forall v \in V$, the solution is a **maximum independent set**, i.e. an MIS of largest cardinality.

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For LPs, LOCAL can simulate any non-signaling distribution

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Consider these random variables:

- X_v is the output of node v
- $X = \sum_{v \in V} X_v$, output of the entire graph

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Key facts

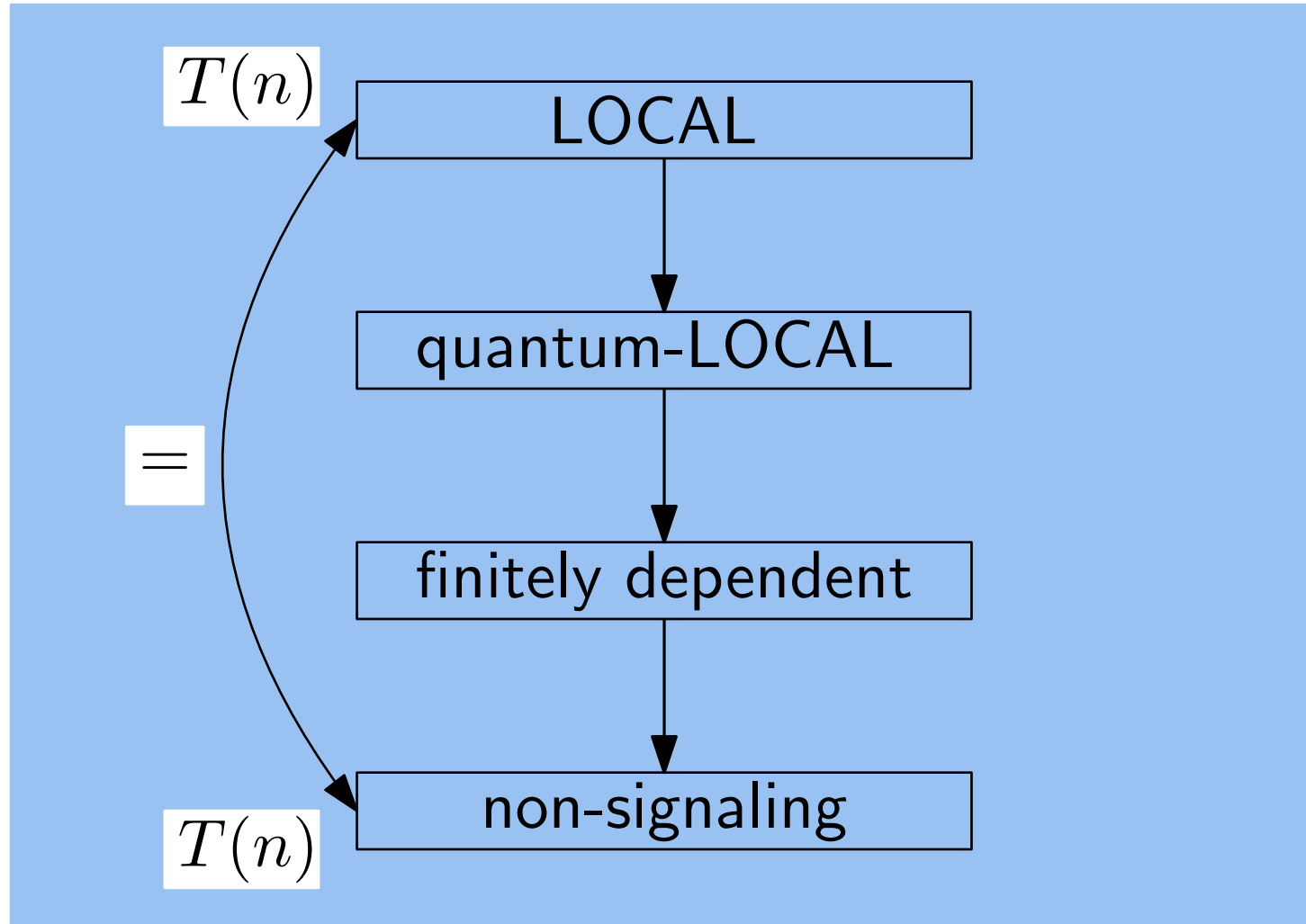
- In $T(n)$ rounds in LOCAL, v can compute $\mathbb{E}[X_v]$
- By linearity of $\mathbb{E}$,

$$\sum_{v \in V} \mathbb{E}[X_v] = \mathbb{E} \left[\sum_{v \in V} X_v \right] = \mathbb{E}[X]$$

- By monotonicity of $\mathbb{E}$, if X is an α -approx. then $\mathbb{E}[X]$ is an α -approx.

Results for Linear Programs (LPs)

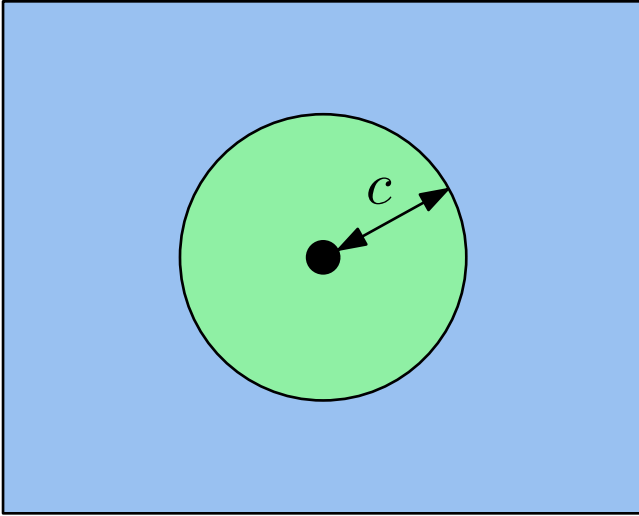
For Linear Programs (LPs)



Results for Locally Checable Labelly Problembs (LCLs)

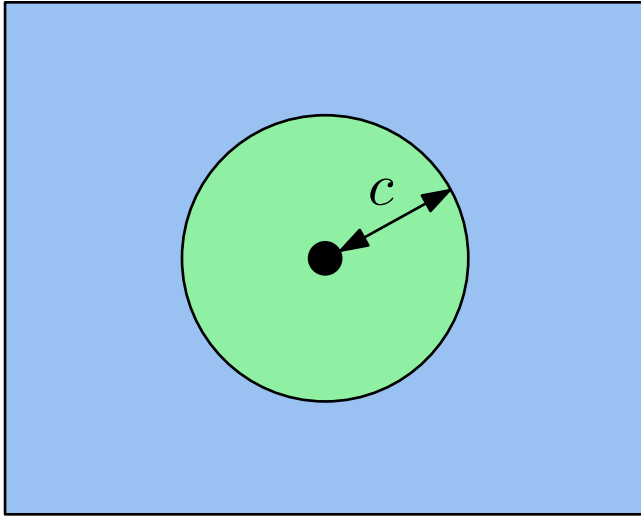
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LCLs: explore up to radius c to check a solution



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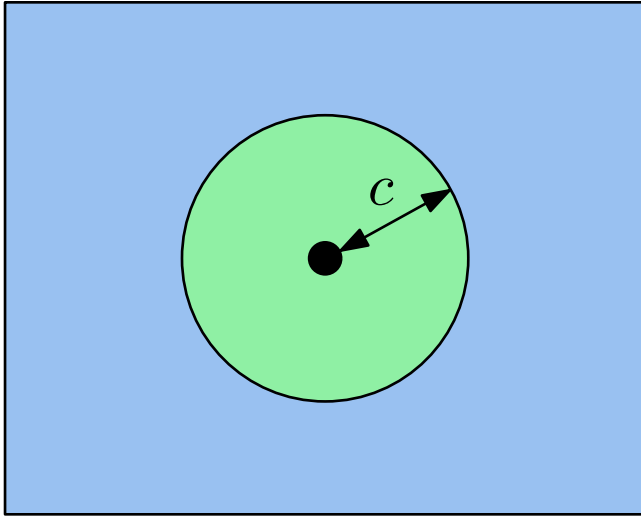


Examples

- MIS, maximal instead of (fractional) maximum
- Coloring
- Maximal Matching
- iterated-GHZ

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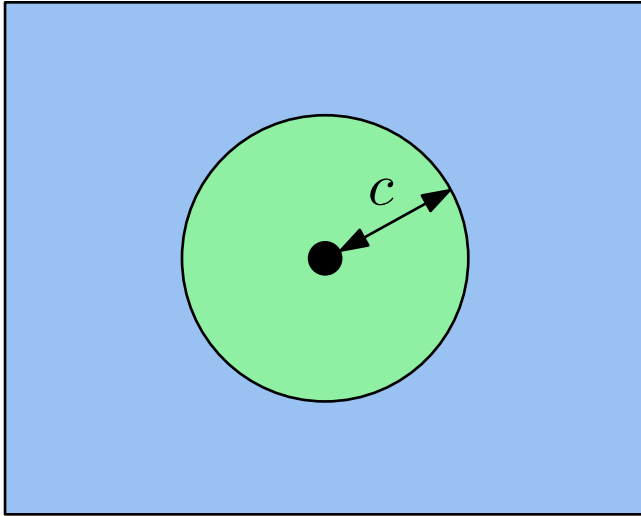
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Reminder on GHZ

Three players A , B and C , and one referee

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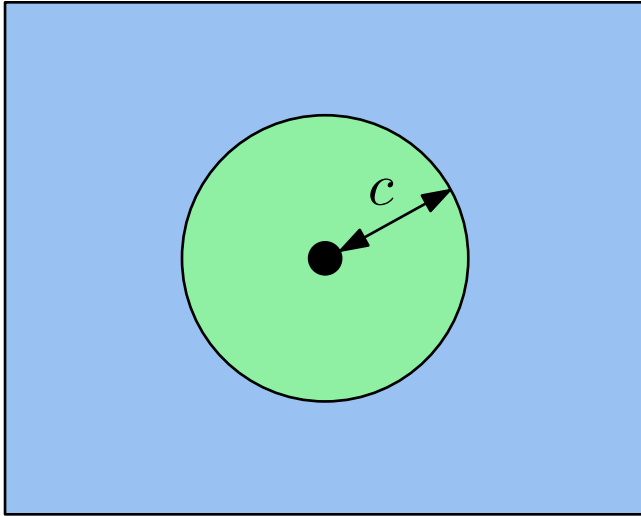
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inputs x , y and z ; outputs: a , b and c

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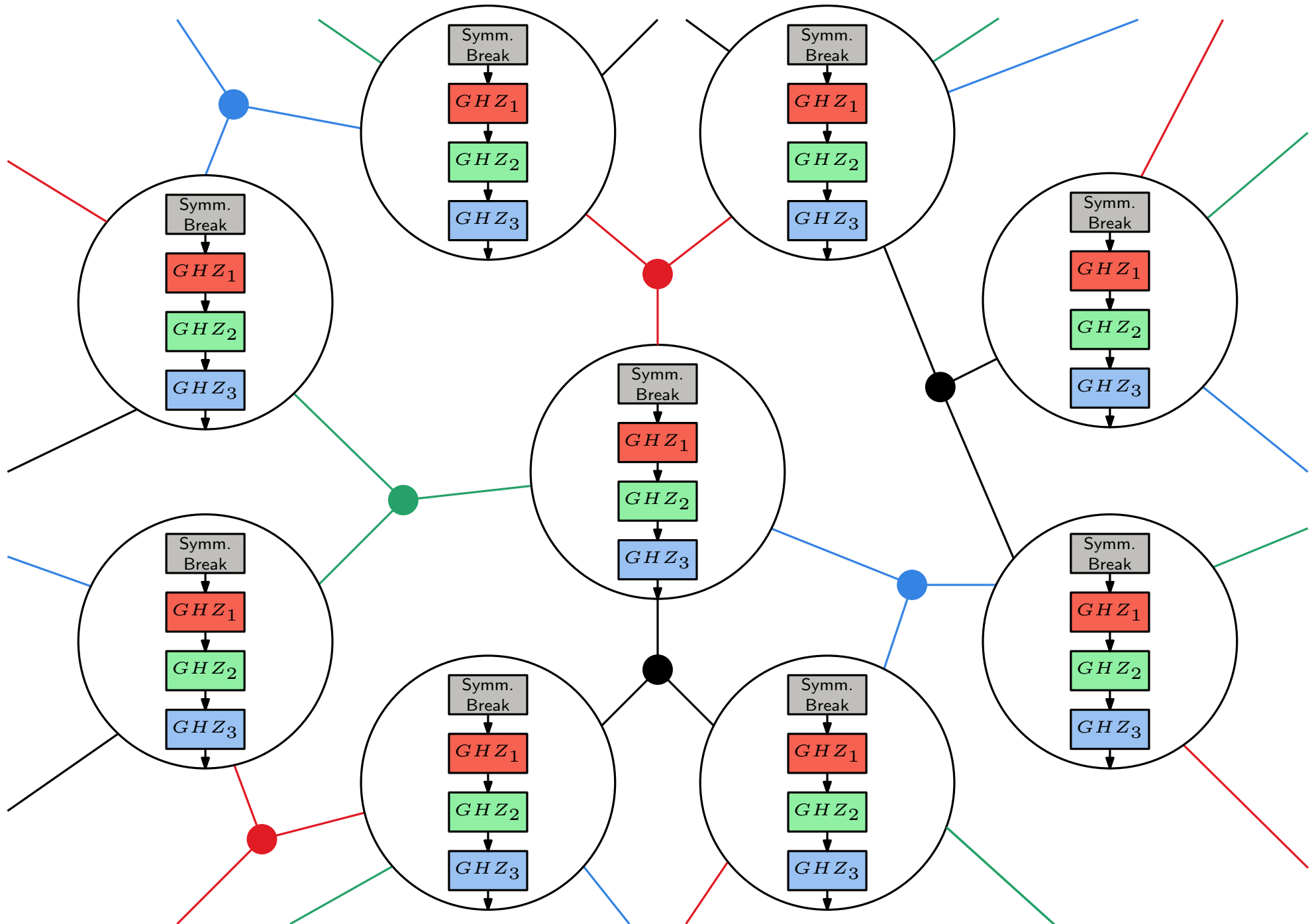
inputs x , y and z ; outputs: a , b and c

Goal: **without communicating after the input is given**, output bits such that

$$a \oplus b \oplus c = x \vee y \vee z$$

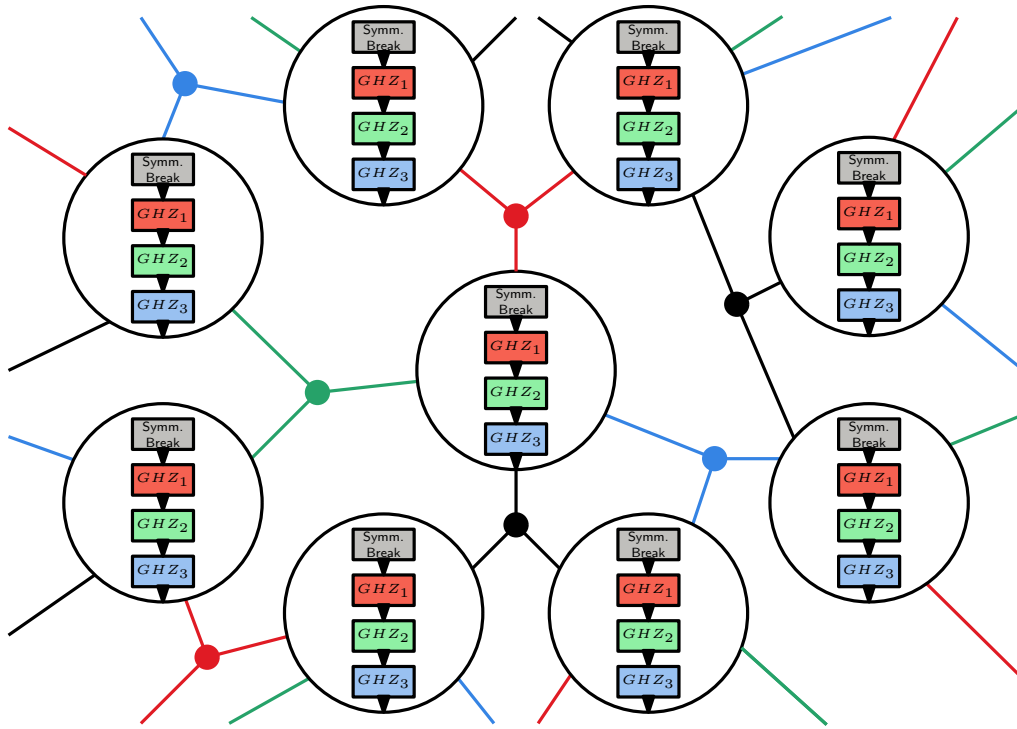
knowing that $x \vee y \vee z \in \{(0, 0, 0), (1, 1, 0), (1, 0, 1), (0, 1, 1)\}$

Results for Locally Checable Labelly Problembs (LCLs)



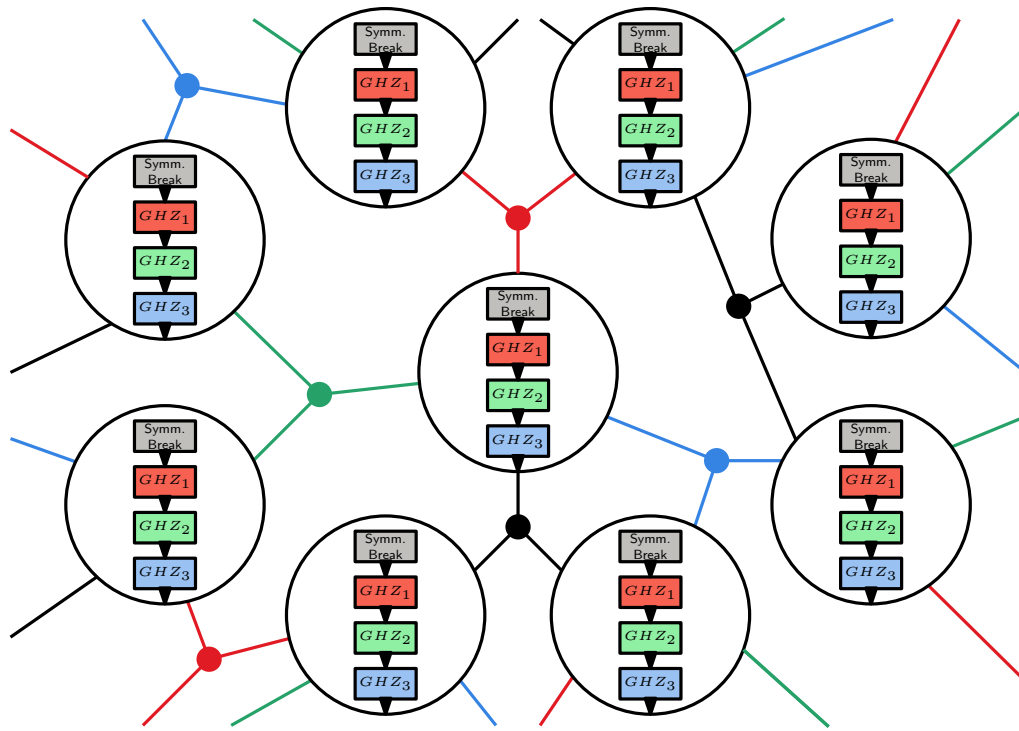
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Round complexity of iterated-GHZ



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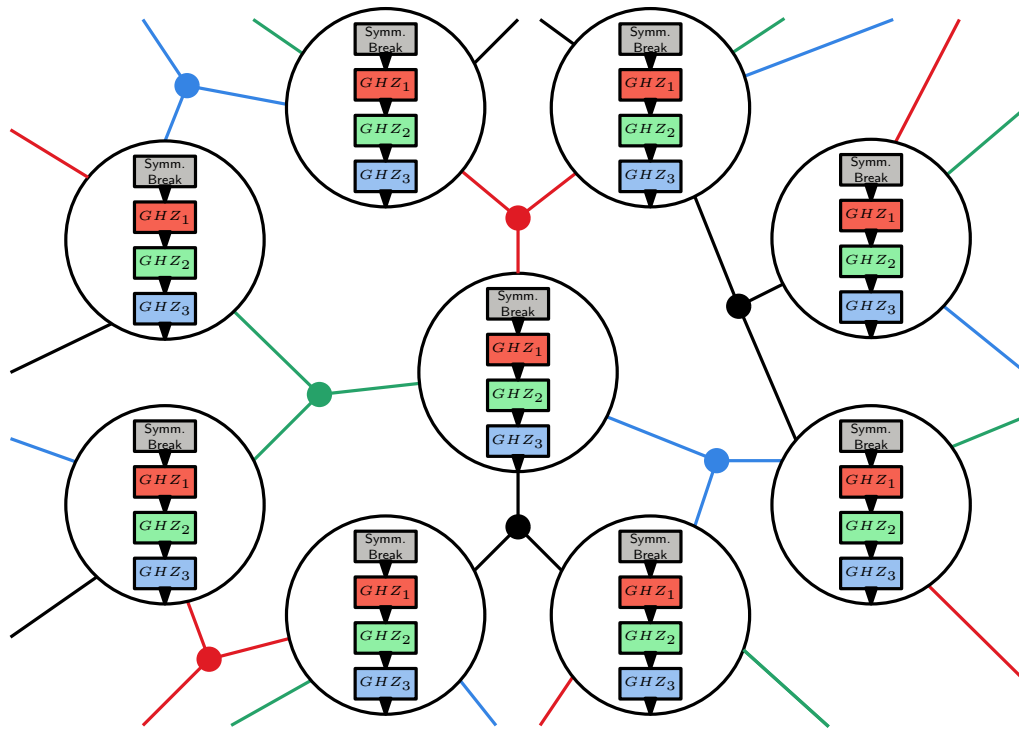
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quantum-LOCAL: $O(1)$ rounds

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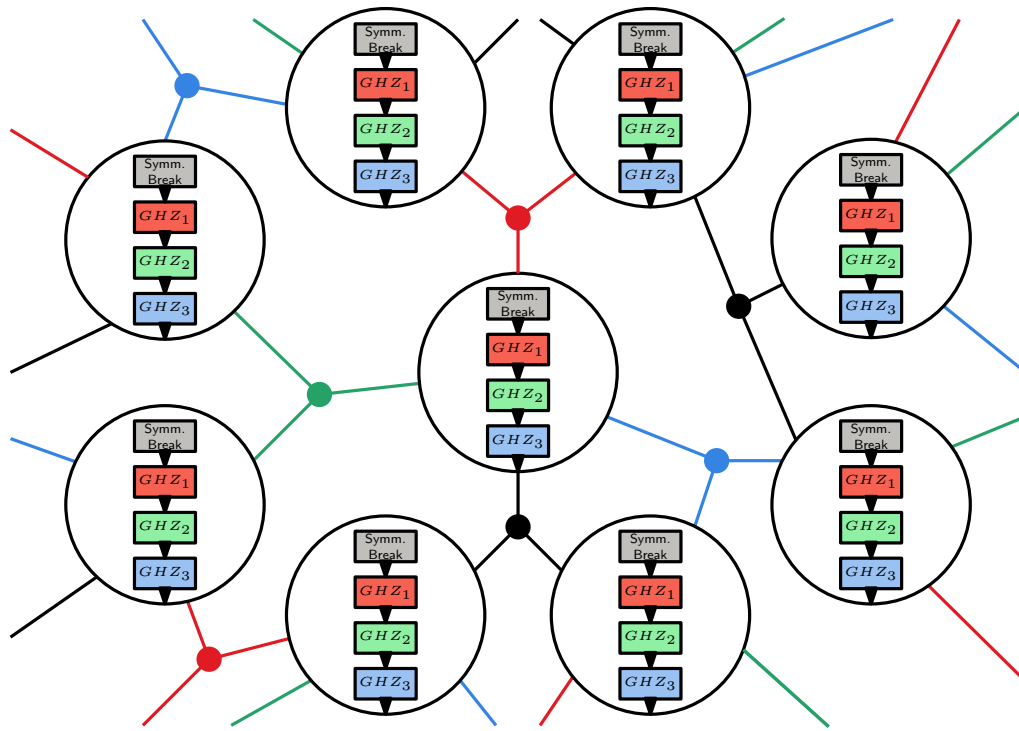


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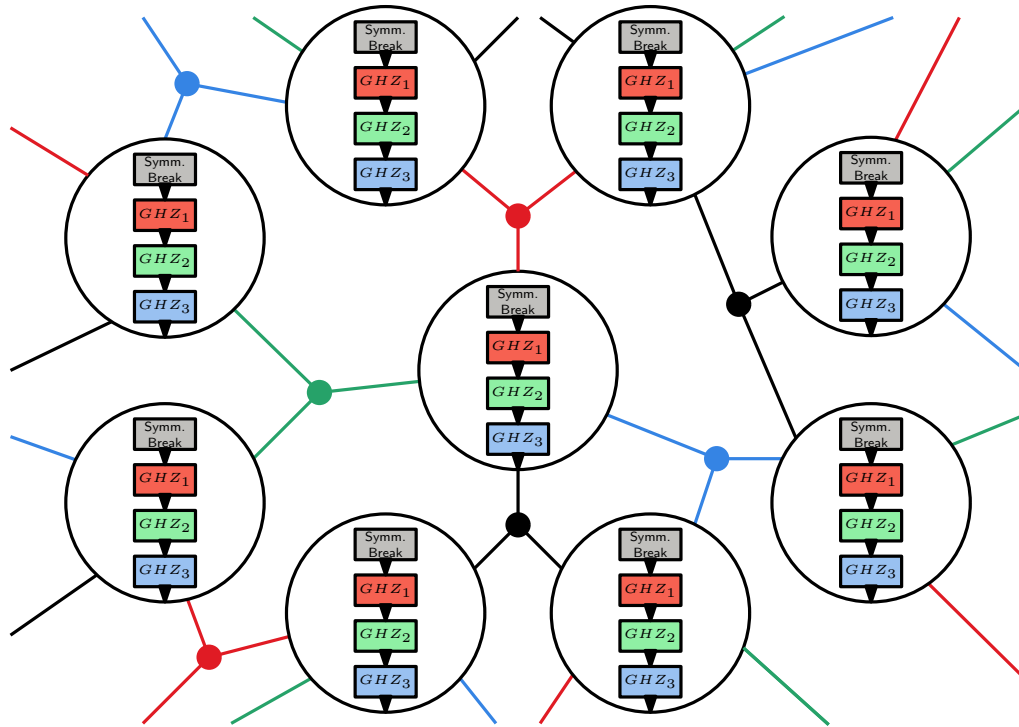


quantum-LOCAL: $O(1)$ rounds

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- measure

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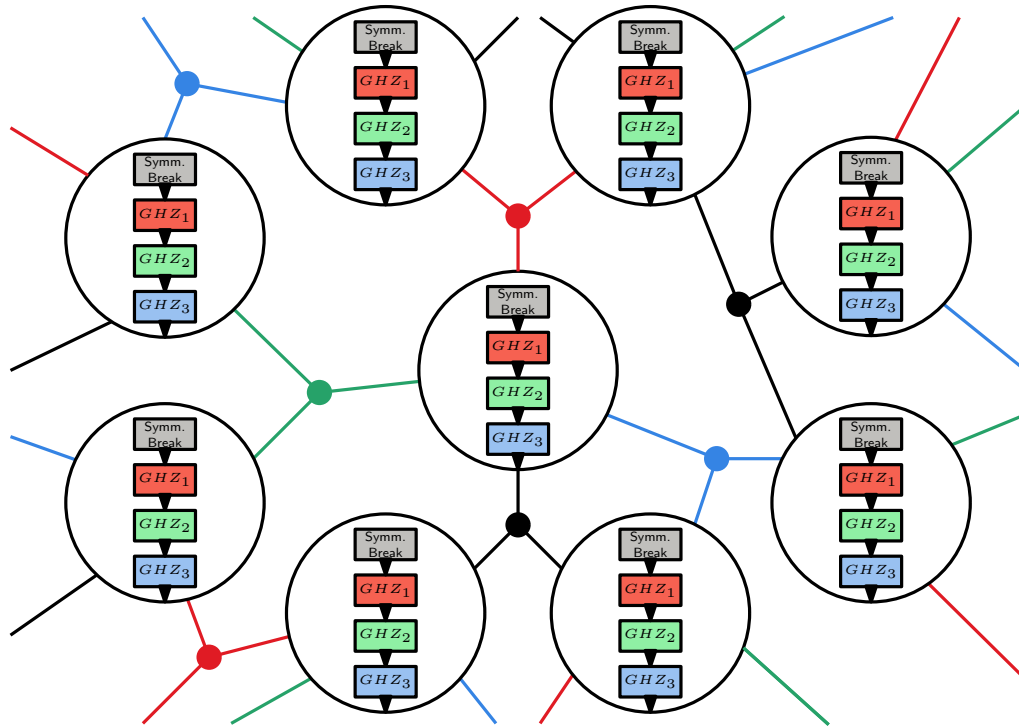
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LOCAL: $O(\Delta)$ rounds

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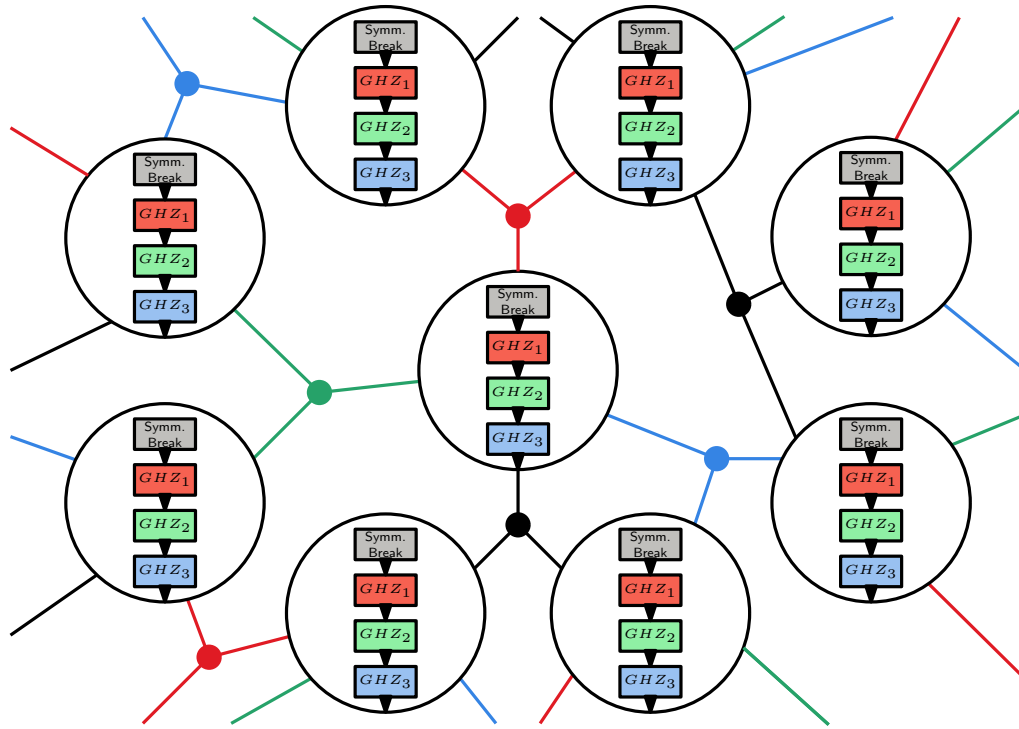
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LOCAL: $O(\Delta)$ rounds

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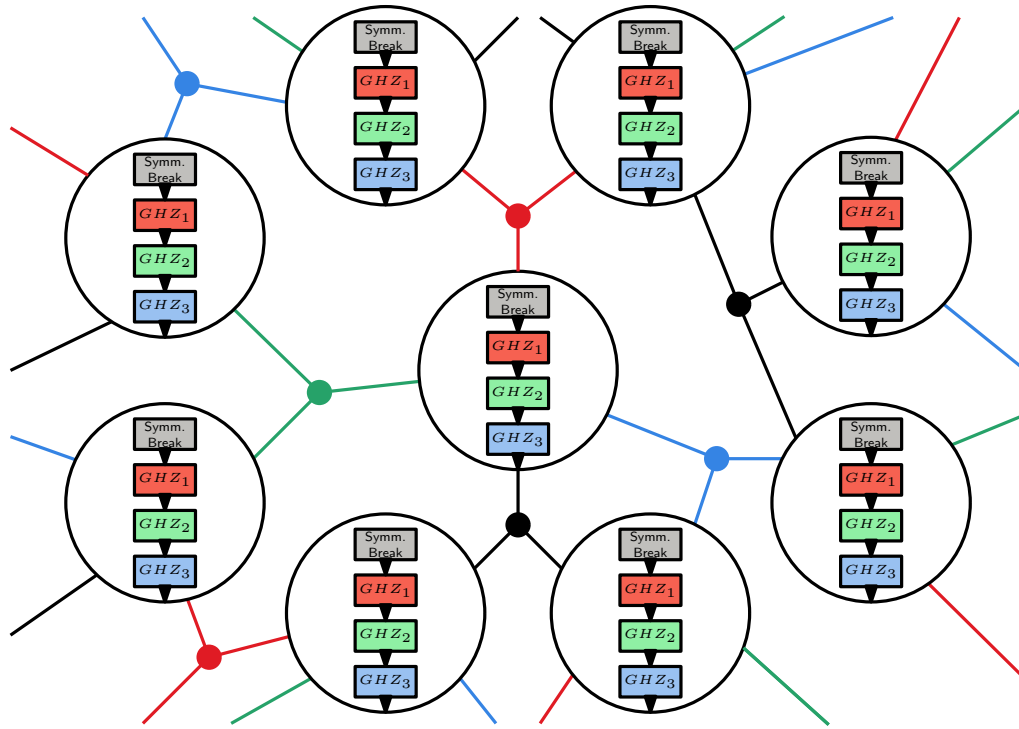
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- solve symm. break. with 1st neigh.
- solve GHZ_1 game with 2nd neigh.

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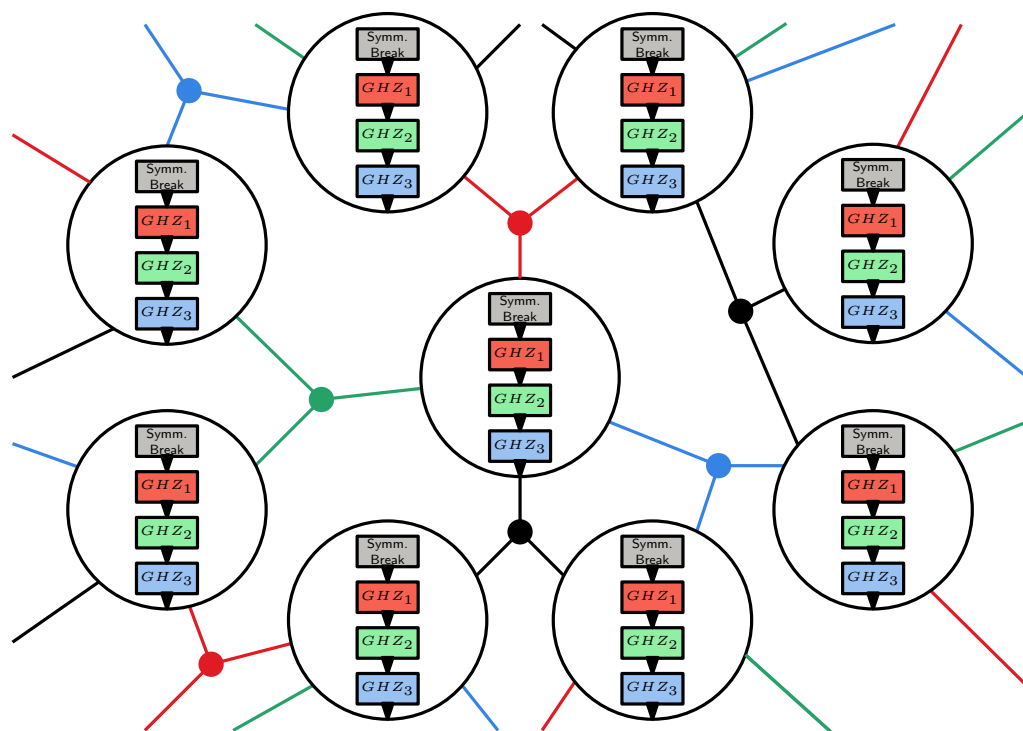
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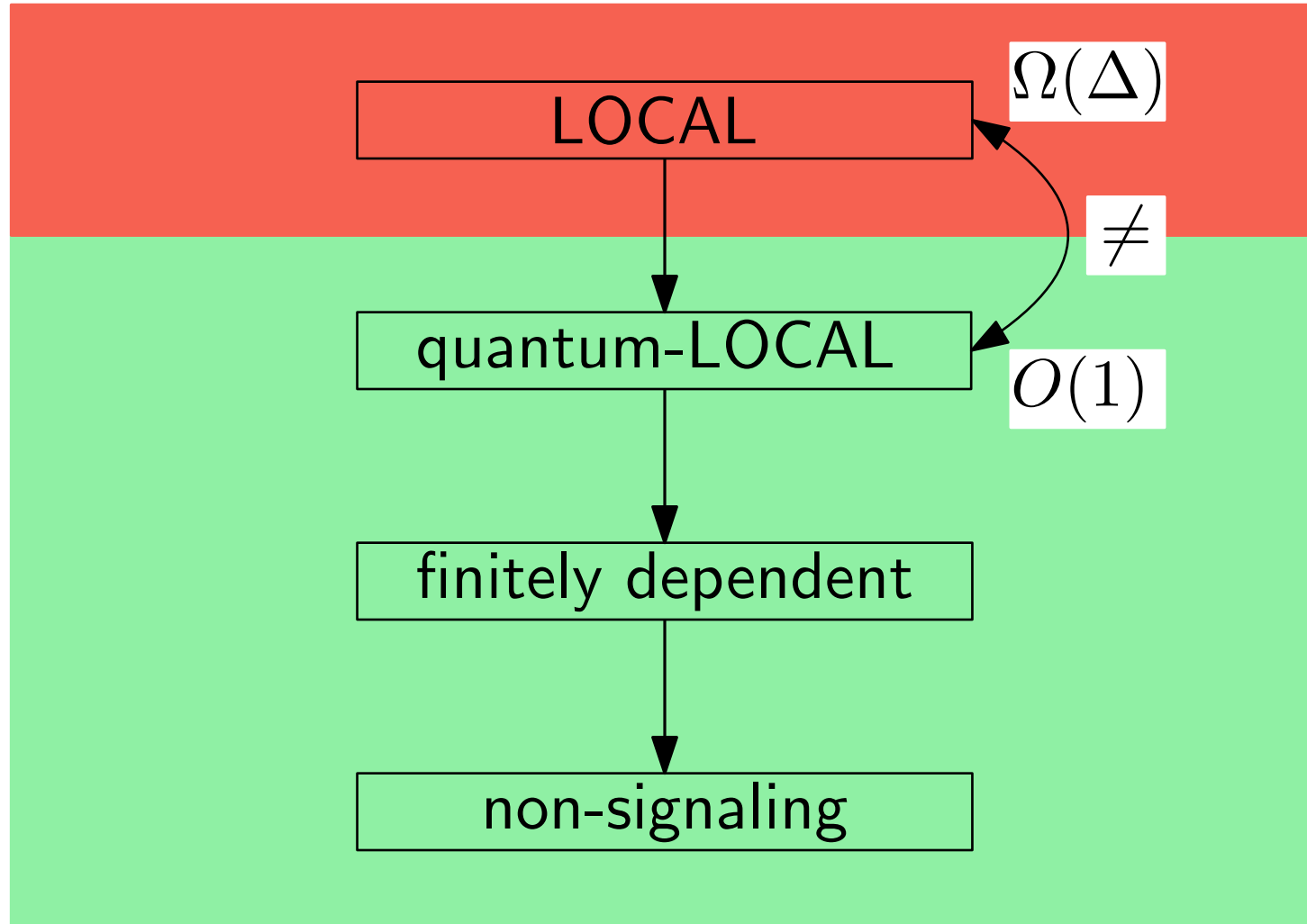
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- solve GHZ_1 game with 2nd neigh.
- solve GHZ_2 game with 3rd neigh.
- ...
- solve GHZ_Δ game with the last neigh.

Results for Locally Checkable Labelling Problems (LCLs)

For Locally Checkable Labelling Problems (LCLs)



Summary

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Linear Programs (also SDPs) : no separation

New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs

A. Balliu, C. Coupet, A. Cruciani, F. D'Amore, M. Equi, H. Lievonen, A. Modanese, D. Olivetti, J. Suomela

DISC 2025

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LCLs : $O(\Delta)$ LOCAL vs $O(1)$ quantum-LOCAL

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STOC 2025

... later improved in SODA 2026

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- only one previous result for LOCAL, and not for LCLs
- most previous literature focused on other models like CONGEST

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Quantum String Matching

From Bit-Parallelism to Quantum String Matching for Labelled Graphs

M. Equi, H. Lievonen, A. Meijer-van de Griend, V. Mäkinen

CPM 2023

Quantum Pattern Matching in Generalised Degenerate Strings

M. Equi, H. Lievonen, Md R. Islam Khan, V. Mäkinen

arXiv 2026, under submission

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Thank you! Questions?